



Figure 2: Consistency for forward multiple-choice test queries by editor and category (dashed line indicates chance).

in their generalizations (Table 3). FT performs the worst, with consistency no better than chance. The consistency of ROME and ICE’s edits are well-above chance, but their performance is roughly half of that for invariant properties, indicating that they fail to fully edit the LLM’s knowledge of a subject’s properties. This finding establishes a clear gap in the performance of these methods, demanding development of more consistent model editors.

Atypical subjects are easier to edit than typical subjects Our results in Table 3 bolster and expand recent works (Ma et al., 2024), where editors were found to perform better for rarer knowledge.

Consistency is consistent across superordinate categories To test the generalizability of findings from TAXI, we also examine editor consistency individually for each of the superordinate categories. We find that editor consistency is similar across categories (Figure 2, right).

5 Human Study

TAXI aims to leverage taxonomic properties to create a clear and intuitive test of consistency for knowledge editors. We validate this with a human study, confirming that TAXI is human-solvable.

In our study, 19 participants (12 Female, 1 non-binary, median age 34 y/o, all in the United States) recruited from CloudResearch’s Connect platform and completed a multiple choice questionnaire analogous to the task used to test language models. A random subset of edits was sampled from TAXI (with one of each exemplar type per category). Each annotator judged 100 items sampled from this subset, for a total of 1,900 human judgments.

For each query, participants were instructed to Imagine a <subject> was a kind of <category>. They were then prompted with the subject and a multiple choice question asking

which of a set of properties applies to the subject. This task is identical to that used to evaluate the editors, so results are directly comparable.

Human annotators dramatically outperform editors Human subjects are approximately *twice as consistent* as the best model editor on the same edits (Figure 2, left), answering correctly on 86.8% of trials (the best-performing participant responded correctly on 95/100 trials). Overall, human behavioral data indicate that the task prescribed by TAXI is human solvable and set a benchmark that far exceeds any existing editors’ performance. Further procedural details are available in Appendix C.

6 Conclusions

We introduce TAXI, a new dataset for evaluating knowledge editors ability to consistently and coherently edit large language models. TAXI is interpretable, building on taxonomic categories, and is designed to evaluate a knowledge edit’s impacts on entailed information. We then propose and study *consistency*, a new metric that measures whether entailed properties are correctly edited, despite an editor never seeing the entailed information. In experiments with recent knowledge editors on Llama-2, we find that consistency varies substantially across existing editors. In editing a subject’s category, we find that the editors preserve existing properties of subjects, while two editors achieve non-trivial consistency. However, human subjects are nearly twice as accurate on the same task, establishing consistent model editing as a new research direction. Overall, TAXI is a challenging, new benchmark for model editors that highlights substantial gaps of existing editing methods. Nevertheless, the fact that existing methods do achieve above-chance performance demonstrates the in-principle feasibility of consistent model editing.

Ethical Considerations

Successful and consistent model editors stand to serve users of artificial intelligence systems in many ways. For instance, editing aspires to improve factuality, reduce harmful LLM generations, support privacy, and potentially reduce costly and environmentally-impactful training requirements. At the same time, unsuccessful or inconsistent model editing for factuality and safety risks instilling false confidence for developers and users. Therefore, stringent evaluations are a key component of editor development. While we aim to support these evaluations with TAXI, no single benchmark is sufficiently comprehensive to ensure consistency of model editing. We urge developers and researchers to adopt TAXI in their evaluations, but we also advocate for the development of further tests and benchmarks. Our human annotation study was conducted under approval from the Institutional Review Board at Arizona State University (IRB approval: 00013322).

Limitations

The categories, subjects, and properties included in TAXI were manually selected, and are likely not entirely representative of these categories, subjects, and properties in natural language. Similarly, TAXI includes only concrete and everyday object categories. It is unclear how editors would perform for more obscure or abstract categories. At the same time, our method for creating TAXI presents a blueprint for the creation of benchmarks that might explore other aspects of editors' performance.

Acknowledgements

We thank Research Computing at Arizona State University for providing High-Performance Computing resources that have contributed to our findings (Jennewein et al., 2023).

References

- Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. 2023. [The Reversal Curse: LLMs trained on "A is B" fail to learn "B is A"](#).
- Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. 2024. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 12:283–298.
- Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. [Knowledge Neurons in Pretrained Transformers](#).
- Nicola De Cao, Wilker Aziz, and Ivan Titov. 2021. [Editing Factual Knowledge in Language Models](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6491–6506, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Tobias Gerstenberg and Joshua B. Tenenbaum. 2017. [Intuitive theories](#). In *The Oxford Handbook of Causal Reasoning*, Oxford Library of Psychology, pages 515–547. Oxford University Press, New York, NY, US.
- Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. 2023. Aging with grace: Lifelong model editing with discrete key-value adapters. In *Advances in Neural Information Processing Systems*.
- Peter Hase, Mona Diab, Asli Celikyilmaz, Xian Li, Zornitsa Kozareva, Veselin Stoyanov, Mohit Bansal, and Srinivasan Iyer. 2023. Methods for Measuring, Updating, and Visualizing Factual Beliefs in Language Models. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2714–2731, Dubrovnik, Croatia. Association for Computational Linguistics.
- Jason Hoelscher-Obermaier, Julia Persson, Esben Kran, Ioannis Konstas, and Fazl Barez. 2023. Detecting edit failures in large language models: An improved specificity benchmark. pages 11548–11559.
- Keith J Holyoak and Dan Simon. 1999. Bidirectional Reasoning in Decision Making by Constraint Satisfaction. *Journal of Experimental Psychology: General*, page 29.
- Zeyu Huang, Yikang Shen, Xiaofeng Zhang, Jie Zhou, Wenge Rong, and Zhang Xiong. 2023. Transformer-patcher: One mistake worth one neuron. In *International Conference on Learning Representations*.
- Douglas M. Jennewein, Johnathan Lee, Chris Kurtz, Will Dizon, Ian Shaeffer, Alan Chapman, Alejandro Chiquete, Josh Burks, Amber Carlson, Natalie Mason, Arhat Kobwala, Thirugnanam Jagadeesan, Praful Barghav, Torey Battelle, Rebecca Belshe, Debra McCaffrey, Marisa Brazil, Chaitanya Inumella, Kirby Kuznia, Jade Buzinski, Sean Dudley, Dhruvil Shah, Gil Speyer, and Jason Yalim. 2023. [The Sol Supercomputer at Arizona State University](#). In *Practice and Experience in Advanced Research Computing*, PEARC '23, pages 296–301, New York, NY, USA. Association for Computing Machinery.
- Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. 2023. [Large Language Models Struggle to Learn Long-Tail Knowledge](#).
- Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. [Infini-gram: Scaling unbounded n-gram language models to a trillion tokens](#).

- Xinbei Ma, Tianjie Ju, Jiyang Qiu, Zhuosheng Zhang, Hai Zhao, Lifeng Liu, and Yulong Wang. 2024. [Is it Possible to Edit Large Language Models Robustly?](#)
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. [When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.
- Vittorio Mazzia, Alessandro Pedrani, Andrea Caciolai, Kay Rottmann, and Davide Bernardi. 2023. A survey on knowledge editing of neural networks. *arXiv preprint arXiv:2310.19704*.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in gpt. In *Advances in Neural Information Processing Systems*.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2023. [Mass-Editing Memory in a Transformer](#). In *International Conference on Learning Representations*.
- Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D Manning. 2022a. Fast model editing at scale. In *International Conference on Learning Representations*.
- Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D. Manning, and Chelsea Finn. 2022b. Memory-Based Model Editing at Scale. In *Proceedings of the 39th International Conference on Machine Learning*, pages 15817–15831. PMLR.
- Yuval Pinter and Michael Elhadad. 2023. [Emptying the Ocean with a Spoon: Should We Edit Models?](#)
- Derek Powell, Kara Weisman, and Ellen M Markman. 2023. Modeling and leveraging intuitive theories to improve vaccine attitudes. *Journal of Experimental Psychology: General*.
- Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, Jennifer Dumas, Yanai Elazar, et al. 2024. Dolma: An open corpus of three trillion tokens for language model pretraining research. *arXiv preprint arXiv:2402.00159*.
- Barbara A. Spellman, Jodie B. Ullman, and Keith J. Holyoak. 1993. [A Coherence Model of Cognitive Consistency: Dynamics of Attitude Change During the Persian Gulf War](#). *Journal of Social Issues*, 49(4):147–165.
- Chenmien Tan, Ge Zhang, and Jie Fu. 2024. Massive editing for large language models via meta learning. In *International Conference on Learning Representations*.
- Paul Thagard. 1989. [Explanatory coherence](#). *Behavioral and Brain Sciences*, 12(3):435–467.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.
- Peng Wang, Ningyu Zhang, Xin Xie, Yunzhi Yao, Bozhong Tian, Mengru Wang, Zekun Xi, Siyuan Cheng, Kangwei Liu, Guozhou Zheng, et al. 2023a. Easyedit: An easy-to-use knowledge editing framework for large language models. *arXiv preprint arXiv:2308.07269*.
- Song Wang, Yaochen Zhu, Haochen Liu, Zaiyi Zheng, Chen Chen, and Jundong Li. 2023b. Knowledge editing for large language models: A survey. *arXiv preprint arXiv:2310.16218*.
- Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang. 2023. [Can We Edit Factual Knowledge by In-Context Learning?](#)
- Zexuan Zhong, Zhengxuan Wu, Christopher D Manning, Christopher Potts, and Danqi Chen. 2023. MQuAKE: Assessing knowledge editing in language models via multi-hop questions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

	FT	ROME	ICE
Edit Success	.05	.10	1.0
Property Success	.05	.12	1.0
Invariance	.10	.21	1.0
Consistency	.04	.10	1.0
Consistency			
Typical Subj.	.05	.09	1.0
Atypical Subj.	.03	.10	1.0

Table 4: Editor and human performance for reverse queries from the TAXI dataset. Edits exhibit stronger invariance than consistency, but both values vary across editing methods. Note that ROME’s edit success was imperfect, suggesting its performance might improve through hyperparameter tuning.

A Example of a TAXI Taxonomy

While all data are publicly available, we also include examples in Table 5, which includes the structure of categories and properties for the superordinate category of animals. The full taxonomy is available in the project repository. Table 2 shows the number of categories and properties for each superordinate category.

Below is a schematic example of a specific edit, property, and associated query and response options.

```
{
  Edit: "A Siamese is a kind of dog."
  Property: "makes sound"
  Forward Query: "A sound a Siamese makes is"
  Responses: ["bark", "chirp", "meow", "moo"]
}
```

B Reverse query performance

We find that reversing queries leads FT and ROME to fail, as shown in Figure 3 and Table 4. This is expected due to the directional nature of "causal" or decoder-only language models like Llama-2 (Touvron et al., 2023), to which we apply these editors. Due to the nature of the causal language model architectures, the effects of model editing methods that seek to edit a specific "subject" are only apparent if the tokening of that subject appears in the context prior to an answer (Berglund et al., 2023; Meng et al., 2022). In contrast, as shown in Figure 3, ICE scores perfectly on the benchmark.

However, we suspect this may reflect ICE’s use of a largely trivial process, whereby the presence of the subject token in the prompt increases its

probability for subsequent generation. One reason for this suspicion is the finding that, for reversed queries, ICE outperforms human annotators performance on forward queries, indicating this performance is likely inflated or meaningless. Further, we speculate that the presence of "reversed" queries in the RIPPLEEDITS benchmark may at least partly explain the relative success of ICE on this benchmark (Cohen et al., 2024).

C Human Study Details

A total of 19 human annotators (12 Female, 1 non-binary, median age 34 y/o, all located in the United States) were recruited from CloudResearch’s Connect platform and asked to complete a multiple choice questionnaire analogous to the task used to test language models. A random subset of edits was sampled from TAXI (with one of each exemplar type per category). Each annotator judged 100 items sampled from this subset, for a total of 1,900 human judgments. Annotator were compensated \$2.25 for their participation, which typically took about 10 minutes.

For each query, annotators were given a prompt to: Imagine a <subject> was a kind of <category>. They were then prompted with the subject and a multiple choice question asking which of a set of properties applies to the subject. Figure 4 displays an example annotation trial.

The full text of instructions given to annotators was:

In this study you will be asked to imagine that different entities belong to different categories. For instance, you might be asked to imagine that "a Parrot is a kind of fish." You should imagine that an entity (e.g. Parrot) inherits the properties of the category to which it belongs (e.g. fish). So if you imagine that "a Parrot is a kind of fish," then you should imagine that a Parrot has scales, swims in the water, and so forth.

On each trial, you will be asked to imagine and to answer a multiple choice question based on the scenario you are imagining. Some of the trials will have different numbers of choices, in which case later options will appear blank. Please ignore any blank option choices.

Annotators agreed with the chosen "correct" answer on 86.8% of trials. The best-performing anno-